

Sam Lokanc

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WEBSITE <https://samlokanc.github.io/>

Relevant Skills

- Python
- SQL
- Data Visualization
- Machine Learning
- R
- Bash / Command Line
- Statistical Analysis
- Geospatial Analysis

Experience

Biodiversity Data Science Research Assistant

University of British Columbia, Vancouver BC

- Provided research assistance to both the MESH (*Multifunctional Ecosystems for Sustainable Human-influenced Landscapes*) and PIEE (*Plant-Insect Ecology and Evolution*) labs.
- Developed **end-to-end data pipelines** leveraging **machine learning** to analyze and present population monitoring data.

May 2026 - Aug 2026

Bioinformatics Research Associate

Arnie Charbonneau Cancer Institute, Calgary AB

- Led an independent bioinformatics research study examining the relationship between cancer and inborn errors of immunity (IEI)-related gene expression.
- Authored a peer-reviewed manuscript published in *Human Immunology*.
<https://doi.org/10.1016/j.humimm.2026.111671>
- Presented findings at multiple academic conferences, earning **2nd place** in a competitive research presentation.

May 2024 - Mar 2026

Student Researcher

Arnie Charbonneau Cancer Institute, Calgary AB

- Selected for a competitive **O'Brien Institute summer research grant** to conduct independent cancer immunotherapy research
- Designed **computational workflows** to support neoantigen-based vaccine development for ependymoma
- Presented and defended findings to academic audiences

May 2023 - May 2024

Projects

Uncovering Opioid Poisoning and Dependence Patterns

Capstone Project - BC Center for Disease Control / University of British Columbia

- Worked in a team to extract useful diagnostic criteria from free-text narrative data using **unsupervised machine learning**.

May 2026 - June 2026

Image Classification of Agricultural Pests

Research Project - Biodiversity Data Science Research Associate Role (PIEE lab)

- Designed an **image segmentation and classification model** to detect agricultural pests. (Funded by Agriculture and Agri-food Canada)

May 2026 - Aug 2026

Bird Acoustic Monitoring Pipeline and Dashboard

Research Project - Biodiversity Data Science Research Associate Role (MESH lab)

- Developed a **bash-based computational pipeline** and **interactive dashboard** to aid population monitoring of Vancouver bird species using acoustic data.

May 2026 - Aug 2026

Gene alterations in inborn errors of immunity and their presence in cancers

Research Project - Bioinformatics Research Associate Role

- Designed and implemented an RNA-seq analysis pipeline using **R and Python** to analyze differential gene expression across cancer and IEI-datasets

May 2024 - Mar 2026

Ependymoma Neoantigen Vaccine Design

Undergraduate Thesis - University of Calgary

- Developed a **bash-based computational pipeline** leveraging **machine learning based prediction tools** to identify neoantigen vaccine candidates

May 2023 - May 2024

Education

Master of Data Science

University of British Columbia, Vancouver, BC

Sept 2025 - Present

Completion June 2026

Bachelors of Health Science in Bioinformatics (First Class Honours)

University of Calgary, Calgary AB

Sept 2020 - May 2024